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# The Clean Core Accelerator

A free, community-built accelerator for SAP Clean Core modernization. It runs a deterministic evidence engine *first*, then AI — turning legacy custom ABAP into Clean-Core-compliant drafts and cryptographically signed audit evidence for architect review and approval. Complementary to SAP's own tooling, never a replacement.

## 01. What is Clean-Core.io?

Clean Core means keeping the standard SAP ERP core untouched. When custom code is mixed directly into standard classes and tables, every future S/4HANA update becomes slow, risky and expensive.

**Clean-Core.io** is a free community tool for SAP practitioners. Upload your legacy ABAP (raw database SELECTs, obsolete function modules, dynpro logic) and it generates the **first Clean-Core-compliant draft plus signed evidence** for review and approval — saving days of manual mapping and boilerplate, without ever replacing the judgment of the expert.

**The governing principle — “belegt, nicht behauptet” (proven, not claimed):** a deterministic ABAP evidence engine runs *before* any AI. Every finding, score and routing decision is tied to concrete evidence in your code. The AI writes the human-readable narrative on top — and that narrative is deliberately excluded from the signed evidence, so a signature always attests to server-computed facts, not to free text.

### THE PROBLEM: Z-TABLE CHAOS & RIGID CORE MODIFICATIONS

Historically grown SAP systems contain hundreds of custom Z-tables and direct modifications. These objects block standard upgrades, because SAP can change the underlying standard structures at any time.

The Clean Core Accelerator analyzes your legacy Z-objects and recommends the optimal modernization track — **In-App ABAP Cloud (RAP)** for tightly-coupled logic, or **Side-by-Side BTP (CAP)** for decoupling — with a transparent, evidence-backed rationale.

**S/4HANA Extensibility Rule:** custom developments belong either as In-App Extensibility in the SAP ABAP Cloud stack (RAP) or as Side-by-Side Extensibility on SAP BTP (CAP). The standard core stays completely clean and upgrade-safe.

## 02. The Value, Part 1: Evidence-First Acceleration

Clean-Core.io is not a code translator — it is an evidence-first Clean Core accelerator. Architects, developers and decision-makers get immediate, defensible advantages:



### EVIDENCE, NOT OPINIONS

A deterministic AST engine parses your code and maps every custom-table access, RFC call and dynpro pattern to concrete findings — *before* any AI runs. You get a defensible baseline you can take into an audit, not a black-box guess.



### DAYS → MINUTES

Manually finding released SAP APIs and rewriting legacy ABAP into RAP classes or CAP services takes days to weeks. Clean-Core.io produces the **first qualified draft plus evidence in minutes** — the expert then reviews, instead of starting from a blank page.



### UPGRADE RESILIENCE

By replacing unreleased database dependencies with officially released SAP APIs (e.g. `I_Customer`, `API_PRODUCT_SRV`), your ERP core stays fully upgradable — decoupling update cycles from months to days.



### AUTOMATED TEST STUBS

Matching unit tests are generated with the code — ABAP Unit test doubles for RAP, or Express/Node test suites for CAP — and executed in an isolated, hermetic sandbox. QA cycles start covered, not empty.



### THE FAST-PASS ADVANTAGE **ACCELERATOR**

A guided 7-stage workflow takes you from Analyze → Design → Transformation → Documentation → Testing → TCO → Delivery. The Fast-Pass Stepper lets you jump straight from analysis to the Delivery view and download a compiled ZIP with all modularized project files, standard abapGit export and tests.

## 02. The Value, Part 2: Transparency & Governance

Version 2.0 makes the reasoning inspectable and the progress auditable — so a recommendation survives scrutiny in a board room, not just a demo.



### “WHY THIS ROUTE & SCORE”

A transparency panel shows exactly why a route (RAP / CAP) and score were chosen: the deterministic router rationale, a confidence indicator, and the driving data-coupling findings by risk. Evidence you can defend — not an AI guess.



### MODULE RISK HEATMAP

A LOC-weighted treemap of the detected ABAP objects grouped by module: tile size = share of the codebase, colour = worst criticality inside it. See at a glance which modules carry the most weight *and* the most risk — where to focus first.



### ARCHITECTURE DECISION RECORD

Every signed evidence pack now includes a Markdown ADR: the decision, the engine recommendation and any architect override, the rationale, considered options, scope & consequences, and known limitations — governance documentation, generated for you.



### RUN-OVER-RUN PROGRESS

Because every analysis is an immutable, signed run, the board deck can show a tamper-evident trend: Clean Core Score, findings and complexity deltas versus the previous run — a defensible record of remediation progress, not a re-editable status slide.

### 3 Scores

CLEAN CORE · COMPLEXITY ·  
CRITICALITY

### BPMN 2.0

+ RACI & LEVEL-5 SOP  
BLUEPRINTS

### TCO & ROI

C-LEVEL UPGRADE-IMPACT  
CALCULATOR

**Portable by design:** outputs are standard — abapGit ZIP export, ABAP-Unit / Express tests, BPMN 2.0 XML (importable into SAP Signavio or SAP Build) and a signed audit pack. You own what you generate. No lock-in.

## 03. Security, Part 1: Your Keys & Your Code

Proprietary legacy source code is a highly confidential business asset. Clean-Core.io is engineered with strict, verifiable boundaries around credentials and AI processing.

### BYOK — SERVER-SIDE AES-256-GCM, NEVER CLIENT-EXPOSED

Bring Your Own Key is optional: without it, every account gets **5 free transformations** on a shared community key; with your own Google Gemini key, usage is unlimited. Your key is encrypted at rest with **AES-256-GCM** in a server-only Firestore store ( `allow read, write: if false` ).

The plaintext key is **never returned to the browser** — only the last four characters are ever shown. It is used exclusively to forward your request to the Gemini API through a secure server-side proxy. No credit card; you pay only for your own Google usage.

### KEYS NEVER REACH THE CLIENT

Every AI call is proxied through a hardened server route: a strict model allowlist, a prompt-size cap, per-user rate limiting, and an MFA gate on sensitive actions. Provider keys never touch client code.

### MODEL-TRAINING ISOLATION

Under Google's applicable Gemini API data-use terms, the code you send is not used to train Google's foundational models. When you use your own key (BYOK), the terms of your own Google account apply. Your IP stays yours.

### TENANT SECURITY ARCHITECTURE

Optional S/4HANA connections are strictly **read-only** (OData metadata reads and test execution) — no write operations. Credentials are processed in memory only and never persisted; production domains are automatically blocked, so only non-productive sandbox/test systems are reachable.

Every live-tenant request is **manually reviewed and approved** by the platform administrator before activation — no unauthorized system access.

**Honest boundary:** for the paid Gemini API the “not used for training” terms apply directly; if you bring a free-tier key, Google's free-tier data-use terms govern instead. We state the applicable terms rather than an absolute promise we cannot control.

## 03. Security, Part 2: Trust Chain & Data Protection

The output is only trustworthy if it is tamper-evident and your data is handled to EU standards. Both are built in.

### IMMUTABLE, SIGNED RUNS

Every analysis is captured as an immutable, HMAC-signed “Run”. The evidence pack is generated and signed **server-side**, so a valid signature attests to server-computed content — closing any client-forged-hash gap.

### THREE-TIER VERIFICATION

Anyone can verify a pack's manifest hash and signature. The AI narrative is deliberately excluded from the signed payload — the signature attests to evidence, not to free text.

### GDPR-ALIGNED DATA PROCESSING & EU SOVEREIGNTY (ART. 17 & 25)

All server, database and compute layers run entirely within the GCP region **Belgium (europe-west1)**, satisfying EU data-residency requirements. Transfers to US sub-processors are safeguarded by EU Standard Contractual Clauses.

Account deletion triggers an atomic erasure routine (Art. 17) that **immediately deletes your live database and authentication entries**; residual copies in encrypted backups age out within 30 days. The system never parses or stores actual business master data.

### APPLICATION HARDENING

A strict Content-Security-Policy, server-side HTML sanitization (DOMPurify), and authenticated, token-verified API routes for every mutating action.

### SUPPLY-CHAIN ASSURANCE

CI secret scanning, a dependency-audit gate and a CycloneDX SBOM on every build; GitHub Actions pinned by commit SHA. Only permissive open-source licenses in the dependency tree.

**Cookies & tracking:** only strictly necessary Firebase Auth storage is used — no analytics, advertising or profiling trackers, and no third-party marketing pixels.

## 04. Technical Details: RAP vs. CAP & API Mapping

During analysis, the engine decides — from syntax and coupling evidence — which extensibility path best fits each object:

| TECHNICAL DIMENSION | IN-APP ABAP CLOUD (RAP)                             | SIDE-BY-SIDE (BTP CAP)                        |
|---------------------|-----------------------------------------------------|-----------------------------------------------|
| Runtime Engine      | Runs natively within the S/4HANA core.              | Runs decoupled on SAP BTP (Node.js/TS).       |
| Interfaces          | Synchronous released CDS views.                     | Decoupled via OData APIs or Event Mesh.       |
| RISE Compliance     | Strict SaaS compliance (zero core modifications).   | Upgrade-resilient Tier-2 custom API wrappers. |
| Focus Case          | Immediate database updates and transactional locks. | Customer portals, mobile apps, external SaaS. |

### AUTOMATED API MAPPING & DECOUPLING

Direct SELECTs on legacy tables (e.g. `KNA1`, `BSEG`) are audit blockers. The engine maps them to released standard interfaces, grounded in SAP's own Apache-2.0 Cloudification Repository:

- ✓ `KNA1` (Customer Master) → CDS View `I_Customer`
- ✓ `BSEG` (Accounting Segment) → CDS View `I_JournalEntry`
- ✓ `MARA` (Material Master) → OData API `API_PRODUCT_SRV`
- ✓ `VBAK` (Sales Header) → RAP Entity `I_SalesOrderTP`

**Built on open data:** the object catalog is grounded in SAP's Apache-2.0-licensed Cloudification Repository, merged with Clean-Core.io's curated mappings. The platform is free to use and built on open standards — the reasoning is transparent, the outputs are portable.

## 05. Honest Scope & How to Join

Trust is built on being clear about the boundaries. Here is what Clean-Core.io deliberately does — and does not — do.

### WHAT IT DOES

- ✓ Generates a first Clean-Core-compliant draft + signed evidence
- ✓ Runs deterministic analysis before any AI
- ✓ Recommends RAP / CAP with a transparent rationale
- ✓ Produces tests, BPMN, TCO and an audit pack

### WHAT IT DOES NOT DO

- ✓ Perform or guarantee an automated migration
- ✓ Replace SAP's own tools (ADT, ATC) or expert judgment
- ✓ Deliver production-ready code without architect review
- ✓ Claim any affiliation with or endorsement by SAP SE

All generated output is a **draft for expert evaluation**. Review, test and approve it with a qualified SAP architect before any productive use. Clean-Core.io is a free, non-commercial community project provided for research and evaluation — independent, and not affiliated with SAP SE or Google LLC.

### JOIN THE COMMUNITY

Generate the first Clean-Core-compliant draft for review and approval. Start with 5 free transformations, or bring your own Gemini API key (BYOK) for unlimited access — no credit card required.

**GET STARTED → [HTTPS://CLEAN-CORE.IO](https://clean-core.io)**

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